

Expected future Reward at state s

Bellman Equation Derivation

$$V_{\pi}(s) = \sum_a \underbrace{\pi(a|s)}_{\text{Policy}} \sum_{s'} \sum_r \underbrace{[r + \gamma V_{\pi}(s')]}_{\text{Discount}} P(s', r | a, s)$$

reward from the env, after the action and state change

Stochastic environments

deterministic

Stochastic probability distribution over actions given the current state

Recursive (DP) $\rightarrow$ Directly solving all possible trajectories from state s not possible hence Dynamic Programming approach.

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

Discounted Rewards

Discount

Determines how much emphasis we want to put on immediate v/s future rewards for a trajectory

mean of these

$$\begin{matrix} G_t^{(1)} \\ G_t^{(2)} \\ G_t^{(3)} \\ \vdots \\ G_t^{(P)} \end{matrix}$$

Discounted Rewards across ALL possible trajectories $1 \rightarrow P$

$$V_{\pi}(s) = E_{\pi} [G_t | S_t = s] \rightarrow \text{Expected future rewards given we are starting at state } s$$

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

$$G_{t+1} = R_{t+2} + \gamma R_{t+3} + \gamma^2 R_{t+4} + \dots$$

$$R_{t+1} + \gamma (R_{t+2} + \gamma R_{t+3} + \dots)$$

$\downarrow$
 G_{t+1}

$$G_t = R_{t+1} + \gamma G_{t+1}$$

$$V_{\pi}(s) = E_{\pi}[G_t | s_t = s]$$

$$V_{\pi}(s) = E_{\pi}[G_{t+1} + R_{t+1} | s_t = s] \quad \because G_t = R_{t+1} + \gamma V_{\pi}(s_{t+1})$$

$$[E(a+b) = E(a) + E(b)] \quad \because \text{Expectation is linear operation}$$

$$V_{\pi}(s) = E_{\pi}(R_{t+1} | s_t = s) + E_{\pi}(\gamma V_{\pi}(s_{t+1}) | s_t = s)$$

$$V_{\pi}(s) = \underbrace{E_{\pi}(R_{t+1} | s_t = s)}_{\text{Expected value of reward I get at the next time step, given I am at State } = s \text{ at current time step}} + \underbrace{\gamma E_{\pi}(V_{\pi}(s_{t+1}) | s_t = s)}_{\text{Discounted Future Reward}}$$

Expected value of reward I get at the next time step, given I am at State = s at current time step

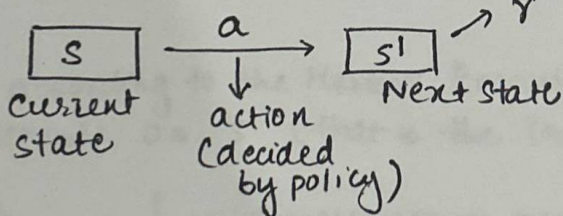
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Immediate Reward

Immediate Reward -

$$E_{\pi}[R_{t+1} | s_t = s] \rightarrow \text{Let's call } R_{t+1} \rightarrow r$$

$$\sum_r r \cdot P(r | s_t = s)$$

A lot of information is hidden on this



So reward is essentially the last step. There are a lot of things that go on on the background to get there

$$P(r | s_t = s) \ni P(r, a, s' | s_t = s)$$

How do we get here?

$$P(r | s_t = s) = \sum_{s'} \sum_a P(r, a, s' | s_t = s)$$

Conditional Probability $\rightarrow$

$$= \sum_{s', a} P(s', r | a, s_t = s) \cdot P(a | s_t = s)$$

$$= \sum_{s', a} P(s', r | a, s) \cdot \pi(a | s) \quad \leftarrow \text{policy}$$

$$E_{\pi}(R_{t+1} | s_t = s) = \sum_r r \cdot \sum_{s', a} P(s', r | a, s) \cdot \pi(a | s)$$

$$= \sum_{r, s', a} r \cdot P(s', r | a, s) \cdot \pi(a | s)$$

Short discourse -

$p(x, y) \rightarrow$ Joint probability dist.

$$\sum_y p(x, y) = p(x)$$

Standard marginalisation

Discounted Future Rewards $\rightarrow$

$$E_{\pi} [G_{t+1} | s_t = s] \Rightarrow \text{Let's call } G_{t+1} \rightarrow g$$

$$\downarrow$$

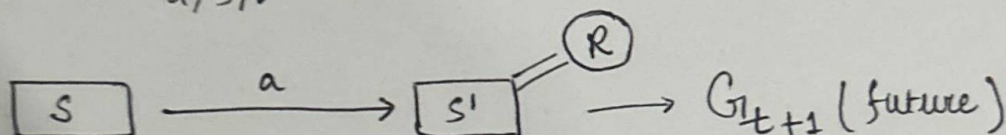
$$\sum_g g \cdot P(g | s_t = s)$$



$$P(g | s) = \sum_{a, s', r} P(g, a, s', r | s)$$

Conditional Probability

$$P(g | s) = \sum_{a, s', r} P(g | r, s', a, s) \cdot P(r, s', a | s)$$



Markov Property :

$$P(X_T | X_{T-1}, X_{T-2}, X_{T-3}, \dots) \triangleq P(X_T | X_{T-1})$$

$\hookrightarrow$ only depends on the previous state

G_{t+1} only depends on s' (that is the last state)

$\hookrightarrow$ SIMPLIFICATION DONE TO MAKE THE MATH TRACTABLE
Assuming the Markov's property $\rightarrow$

$$P(g | s) = \sum_{a, s', r} P(g | r, s', a, s) \cdot P(r, s', a | s)$$

$$= \sum_{a, s', r} P(g | s') \cdot \underbrace{P(r, s', a | s)}$$

We have this from the immediate reward term calculation

$$= \sum_{a, s', r} P(g | s') \cdot P(s', r | a, s) \pi(a | s)$$

$$E[G_t | s_t = s] = \sum_g g P(g | s)$$

$$= \sum_g g \sum_{a, s', r} P(g | s') \cdot P(s', r | a, s) \pi(a | s)$$

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

$$= \underbrace{R_{t+1}}_r + \sum_{k=1}^{\infty} \gamma^k R_{t+k+1}$$

G_{t+1} does not depend on r from current action, but the environment produces next state and reward as a pair, so we marginalise over a, s and r using law of total probability.

$\hookrightarrow$ Check Markov Property below

$$= \sum_{a, s', \gamma} \left[\sum_g g \cdot P(g | s') \right] P(s', \gamma | a, s) \pi(a | s)$$

$$\text{So, } V_{\pi}(s) = \mathbb{E}_{\pi} [G_t | s_t = s]$$

$$= \sum G_t \cdot P(G_t | s_t = s)$$

$$\therefore g = G_{t+1} \quad (\text{as we assumed this earlier})$$

$$\sum_g g \cdot P(g | s') = \mathbb{E}_{\pi} [G_{t+1} | s_t = s'] = \underbrace{V_{\pi}(s')}_{\text{Value function at the next state}}$$

Value function
at the next state

$$\mathbb{E}[G_{t+1} | s_t] = \sum_{a, s', \gamma} V_{\pi}(s') \cdot P(s', \gamma | a, s) \pi(a | s)$$

Putting the two terms together $\rightarrow$

$$V_{\pi}(s) = \mathbb{E}_{\pi} [R_{t+1} | s_t = s] + \gamma \mathbb{E} [G_{t+1} | s_t = s]$$

$$= \sum_{\gamma} \sum_{s'} \sum_{a^*} \underbrace{\gamma P(s', \gamma | a, s) \cdot \pi(a | s)}_{\substack{\cdot P(s', \gamma | a, s) \\ \cdot \pi(a | s)}} + \gamma \sum_{\gamma} \sum_{s'} \sum_a V_{\pi}(s') \cdot \underbrace{P(s', \gamma | a, s) \cdot \pi(a | s)}_{\substack{\cdot P(s', \gamma | a, s) \\ \cdot \pi(a | s)}}$$

$$\sum_{\gamma} \sum_{s'} \sum_a [\gamma + \gamma V_{\pi}(s')] P(s', \gamma | a, s) \cdot \pi(a | s)$$

$$V_{\pi}(s) = \sum_a \pi(a | s) \sum_{\gamma} \sum_{s'} [\gamma + \gamma V_{\pi}(s')] P(s', \gamma | a, s)$$

only depends on a